Braess's Paradox

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Abstract

In this document it explains and experiments with the phenominan reffered to as Braess's Paradox. Testing how systems that work in parallel vs in series can be counterintuitive to standard thought, working with a countersnapping mechanism and weights. As most topics in the universe function on a weight and spring, this paradox can be observed with a large array of topics alike to Hooke's Law: fluids, electron movement, most commonly seen in traffic, and more.

1 Introduction

Braess's Paradox, first proposed by economist Arthur Pigou and later named mathematician Dietrich Braess in 1968, describes a counterintuitive phenomenon in network theory: adding extra capacity to a network has the ability to decrease the overall performance of the system. This occurs when items in the system act "selfishly" to optimize their own utility, leading to a suboptimal **Nash Equilibrium** [Braess, 1968].

1.1 Mechanically

While commonly explained through traffic, I find it easier to explain the system of springs and strings.

Consider a weight supported by two identical springs, each with a spring stiffness (K_1,K_2)and two long pieces of string (T_1, T_2). Initially, the springs are connected in a configuration where they are under high tension. Consider the Hooke's law equations 1 and 2:

$$K_1 + K_2 = K_{parallel} \tag{1}$$

$$\frac{1}{K_1} + \frac{1}{K_2} = \frac{1}{K_{series}} \tag{2}$$

If a new, short connecting string is added between the springs, one might assume it provides more support. However, it can cause the weight to hang lower as the system transferred from in series to in parallel. Conversely, cutting a connecting string can cause the weight to rise, as the load redistribution forces the springs into a parallel configuration [Cohen and Horowitz, 1991].

Similarly, resistors and capacitors in a circuit function differently in series vs in parallel with their units, Ohms and Farads respectively.

1.2 The Mathematical Framework

The paradox is defined by the gap between the Social Optimum (the configuration that minimizes total system cost) and the Nash Equilibrium (where no individual can improve their position by changing strategy alone).

In a network with total flow R, if $c_e(f_e)$ is the cost function of an edge e with flow f_e, the Nash Equilibrium occurs when all used paths P have an equal and minimal cost:

$$\sum_{e \in P} c_e(f_e) = L \tag{3}$$

The Price of Anarchy (PoA) quantifies this inefficiency: When Braess's Paradox occurs, adding an edge increases the cost of the Nash Equilibrium while the Social Optimum remains the same or decreases.

$$PoA = \frac{CostofNashEquilibrium}{CostofSocialOptimum} \tag{4}$$

2 My process

The creator of the mechanism, Paul Ducarme, mentions on his project website that the initial model was cast with silicone in a printed mold. I chose to print the models directly as that allows time in my independant study to have quick prototypes so it doesn't require reprinting and waiting for silicone to cure.

My final countersnapping structure had been printed out of TPU. With more experimental time, flexible resin would have much higher quality and repeatable print though I've had minor success out of it as I've found it to be "spongey" in elasticity rather than "snappy". Perhaps mixing the material with a stiffer resin would make it better for this application.

2.1 Printing

I started out by skimming the supplimentary information paper [Paul Ducarme, 2025] and downloading the countersnapping files from the project website [Ducarme, Paul]. Inside the .zip file contains 3 folders: Connectors, Molds, and Parts. For my build I only used models from the Connectors and Parts directories. From the project site Ducarme directed to create the parts listed:

2 stiffening blocks

1 non-monotonic block

2 softening blocks

2 outer connectors

2 inner connectors

I took listed files and imported the models in my GCode slicer of choice: the open source edition of Ultimaker Cura.

An important note is that it's ideal to print all parts flat as seen in Figure 1. This prevents the layer lines creating weaker links in the parts. Here are some settings in Cura that I've found to be useful in this build:

Layer height: $.16mm$

Bed temp: $70\ ^{\circ}C$

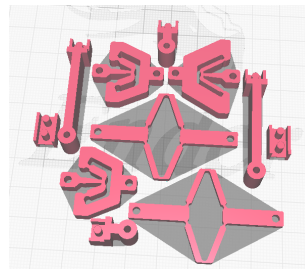

Figure 1: Lineup of all needed files in a 3d printing slicer software.

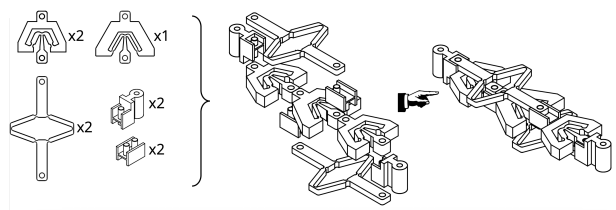

Figure 2: A diagram demonstration on the orientations and positions of each piece of the assembly. [Ducarme, Paul]

Hotend Temp: $210\ ^{\circ}C$

Enable Print Ironing: True

Fan Speed: 50%

2.2 Assembly

I assembled all of the printed parts in reference to Figure 2.

3 Testing Hooke's law edition of Braess' paradox

This mechanism consists of 4 springs. Measured; the 2 outer weak springs have a Hooke's constant (k) of and the inner strong springs have a $k =$. The middle snapping structure is a spring of $k =$. This mechanism would switch having a k_total flipping from a calculated to

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